

Data Structures & Algorithms (DSA) Test — Set A

Difficulty: Easy to Medium

Duration: 45 Minutes

Total Questions: 5

Important Instructions:

- No external references or IDE autocomplete assists are allowed during this assessment.
- Write clean, fully structured, and optimized Java code.
- Pay special attention to standard edge cases (empty collections, single-element collections, boundary limits).

Question 1 — Pattern (Easy)

Write an efficient program to generate and print a **Left-Aligned Inverted Number Triangle** pattern for a given size $n = 5$.

Expected Output:

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

- **Input:** An integer n representing the max rows/columns.
- **Output:** The formatted triangle printed directly to stdout.

Question 2 — String (Easy)

Given a string s , check if it forms a valid palindrome after transforming all uppercase characters into lowercase representations and discarding all non-alphanumeric characters.

- **Example Input:** $s = \text{"A man, a plan, a canal: Panama"}$
- **Example Output:** `true`
- **Explanation:** Processing strings into unified alpha-numerics delivers "amanaplanacanalpanama", which reads identically forward and backward.
- **Constraints:** $1 \leq s.length() \leq 2 \times 10^5$. The input consists exclusively of printable ASCII codes.

Question 3 — Hashing (Easy)

Given an array of integers `nums`, return `true` if any unique value shows up **at least twice** inside the array; return `false` if every single integer element remains strictly unique.

- **Example Input:** `nums = [1, 2, 3, 1]`
- **Example Output:** `true`
- **Constraints:** $1 \leq \text{nums.length} \leq 10^5$; $-10^9 \leq \text{nums}[i] \leq 10^9$.

Question 4 — Array (Easy)

Given a binary array `nums` containing only bits, determine and return the maximum number of consecutive 1s occurring sequentially inside the array structure.

- **Example Input:** `nums = [1, 1, 0, 1, 1, 1]`
- **Example Output:** `3`
- **Constraints:** $1 \leq \text{nums.length} \leq 10^5$; `nums[i]` will strictly be either 0 or 1.

Question 5 — Matrix (Easy)

Given a square matrix `mat` of dimension $n \times n$, accurately calculate and return the total sum of the matrix diagonals. Include all elements situated along the primary diagonal and the secondary diagonal. Crucially, any intersecting elements occupying both diagonals must only be tallied a single time.

Example Input:

```
[[1, 2, 3],  
 [4, 5, 6],  
 [7, 8, 9]]
```

- **Expected Output:** `25`
- **Explanation:** Diagonals addition: $1 + 5 + 9 + 3 + 7 = 25$. Note that center element `mat[1][1] = 5` is added only once.
- **Constraints:** $n == \text{mat.length} == \text{mat}[i].\text{length}$; $1 \leq n \leq 100$; $1 \leq \text{mat}[i][j] \leq 100$.